

No difference when replaying message:

Publisher-

```
[CONNECTING] localhost:8883...
[SUCCESS] Connected to broker as HYDROLOGIC-Device-001
[INFO] Certificate identity verified by broker

[PUBLISHING] Sending sensor readings (Ctrl+C to stop)...

[1] Published: 46.78 LPM
[PUBLISHED] Message ID: 1
[2] Published: 46.94 LPM
[PUBLISHED] Message ID: 2
[3] Published: 51.09 LPM
[PUBLISHED] Message ID: 3
[4] Published: 51.57 LPM
[PUBLISHED] Message ID: 4
[5] Published: 49.78 LPM
[PUBLISHED] Message ID: 5
[6] Published: 48.75 LPM
[PUBLISHED] Message ID: 6
[7] Published: 53.16 LPM
[PUBLISHED] Message ID: 7
[8] Published: 54.57 LPM
[PUBLISHED] Message ID: 8
[9] Published: 54.35 LPM
[PUBLISHED] Message ID: 9
[10] Published: 48.75 LPM
[PUBLISHED] Message ID: 10
[11] Published: 48.8 LPM
[PUBLISHED] Message ID: 11
[12] Published: 53.15 LPM
[PUBLISHED] Message ID: 12
[13] Published: 50.92 LPM
[PUBLISHED] Message ID: 13
[14] Published: 46.23 LPM
[PUBLISHED] Message ID: 14
[15] Published: 49.21 LPM
[PUBLISHED] Message ID: 15
[16] Published: 46.71 LPM
[PUBLISHED] Message ID: 16
[17] Published: 46.64 LPM
[PUBLISHED] Message ID: 17
[18] Published: 49.36 LPM
[PUBLISHED] Message ID: 18
[19] Published: 54.74 LPM
[PUBLISHED] Message ID: 19
[20] Published: 50.87 LPM
[PUBLISHED] Message ID: 20
[21] Published: 46.62 LPM
[PUBLISHED] Message ID: 21
[22] Published: 51.64 LPM
[PUBLISHED] Message ID: 22
[23] Published: 49.13 LPM
[PUBLISHED] Message ID: 23
^C

[INFO] Stopping after 23 messages...
[INFO] Clean disconnect
[INFO] Disconnected from broker
kaylacolletti@Mac hydroficient-project %
```

Subscriber-

```
=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 13:36:17
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 49.13 LPM
Pressure (Up): 58.49 PSI
Pressure (Down): 55.13 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 13:36:32
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 50.92 LPM
Pressure (Up): 59.36 PSI
Pressure (Down): 57.98 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 13:36:33
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 46.23 LPM
Pressure (Up): 59.48 PSI
Pressure (Down): 54.69 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 13:36:34
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 49.21 LPM
Pressure (Up): 58.75 PSI
Pressure (Down): 55.6 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 13:36:35
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 46.71 LPM
Pressure (Up): 60.27 PSI
Pressure (Down): 54.93 PSI
Status: operational

=====
[RECEIVED] Topic: hydroficient/grandmarina/device-001/sensors
[TIME] 13:36:36
=====
Device: HYDROLOGIC-Device-001
Flow Rate: 46.64 LPM
Pressure (Up): 59.37 PSI
Pressure (Down): 56.75 PSI
Status: operational
█
```

Replay-

```
kaylacolletti@Mac hydroficient-project % python3 replay_attacker.py --mode capture --count 5
=====
REPLAY ATTACK TOOL - Capture Mode
=====
Target: 5 messages
Output: captured_messages.json
=====
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/replay_attacker.py:114: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="replay-attacker-capture", **MQTT_CLIENT_ARGS)

[CONNECTING] localhost:8883...
[LISTENING] Waiting for messages to capture...

[CONNECTED] Listening on: hydroficient/grandmarina/#
[CAPTURED 1/5] HYDROLOGIC-Device-001 - 58.92 LPM
[CAPTURED 2/5] HYDROLOGIC-Device-001 - 46.23 LPM
[CAPTURED 3/5] HYDROLOGIC-Device-001 - 49.21 LPM
[CAPTURED 4/5] HYDROLOGIC-Device-001 - 46.71 LPM
[CAPTURED 5/5] HYDROLOGIC-Device-001 - 46.64 LPM

[DONE] Captured 5 messages
[SAVED] captured_messages.json
[SAVED] 5 messages saved to captured_messages.json
kaylacolletti@Mac hydroficient-project % python3 replay_attacker.py --mode replay
=====
REPLAY ATTACK TOOL - Replay Mode
=====
Loaded 5 captured messages
=====
/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/replay_attacker.py:175: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="replay-attacker", **MQTT_CLIENT_ARGS)

[CONNECTING] localhost:8883...

[REPLAYING] Sending captured messages...

[REPLAYING 1/5] Topic: hydroficient/grandmarina/device-001/sensors
  Originally captured: 2026-04-30T17:35:27.038705Z
  Replayed at: 2026-04-30T17:36:32.559099Z

[REPLAYING 2/5] Topic: hydroficient/grandmarina/device-001/sensors
  Originally captured: 2026-04-30T17:35:32.043097Z
  Replayed at: 2026-04-30T17:36:33.564809Z

[REPLAYING 3/5] Topic: hydroficient/grandmarina/device-001/sensors
  Originally captured: 2026-04-30T17:35:37.048465Z
  Replayed at: 2026-04-30T17:36:34.565674Z

[REPLAYING 4/5] Topic: hydroficient/grandmarina/device-001/sensors
  Originally captured: 2026-04-30T17:35:42.052325Z
  Replayed at: 2026-04-30T17:36:35.571296Z

[REPLAYING 5/5] Topic: hydroficient/grandmarina/device-001/sensors
  Originally captured: 2026-04-30T17:35:47.058431Z
  Replayed at: 2026-04-30T17:36:36.572723Z

[DONE] Replayed 5 messages
[NOTE] Check your subscriber - did it accept them all?
kaylacolletti@Mac hydroficient-project %
```